

Design and Analysis of Algorithms a.k.a. Analysis of Algorithms
Special Section 2024 (Summer 24)
Quiz 02 - 25-07-2024

[1 20 marks] Show using a recursion tree method that the recurrence $T(n) = 4T(n/3) + n$ has the solution $T(n) = \Theta(n^{\log_3 4})$.

[2 80 marks] Using the formula of simpler form of Master Method (given at end), obtain the running time of the following recurrences.

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|----|------------------------|------------------|
| 1. | $T(n) = 2T(n/2) + 1$ | Note: 1 is n^0 |
| 2. | $T(n) = 2T(n/2) + n$ | Note: n is n^1 |
| 3. | $T(n) = 2T(n/2) + n^2$ | |
| 4. | $T(n) = 2T(n/2) + n^3$ | |
| 5. | $T(n) = 2T(n/2) + n^4$ | |
| 6. | $T(n) = 3T(n/5) + n$ | |
| 7. | $T(n) = 5T(n/3) + n^2$ | |
| 8. | $T(n) = T(n/2) + n$ | |

SIMPLER FORM OF MASTER METHOD

For recurrences in the form $T(n) = aT(n/b) + \Theta(n^i)$ where

- $a \geq 1$ is the number of subproblems,
- $b > 1$ is the factor by which the problem size is divided in each subproblem,
- $i \geq 0$ is a constant integer representing the degree of the polynomial n^i .

Solution of the above (boxed) recurrence is

- $T(n) = \Theta(n^{\log_b a})$ if $a > b^i$
- $T(n) = \Theta(n^i \log_b n)$ if $a = b^i$
- $T(n) = \Theta(n^i)$ if $a < b^i$